

TITLE: Static Timing Analysis

Objective :

The objective of this project is to design and simulate a Verilog-based digital circuit in EDA Playground, analyze its timing behavior, identify timing violations, and improve performance using timing optimization techniques.

Design Code:

```
`timescale 1ns/1ps

module pipeline_adder(
    input clk,
    input [7:0] a,
    input [7:0] b,
    output reg [8:0] y
);

reg [7:0] a_reg;
reg [7:0] b_reg;

always @(posedge clk) begin
    a_reg <= a;
    b_reg <= b;
    y <= a_reg + b_reg;
end

endmodule
```

Testbench Code:

```
`timescale 1ns/1ps

module tb;

reg clk;
```

```
reg [7:0] a;  
reg [7:0] b;  
wire [8:0] y;
```

```
pipeline_adder dut (  
    .clk(clk),  
    .a(a),  
    .b(b),  
    .y(y)  
);
```

```
always #5 clk = ~clk;
```

```
initial begin  
    $dumpfile("dump.vcd");  
    $dumpvars(0, tb);
```

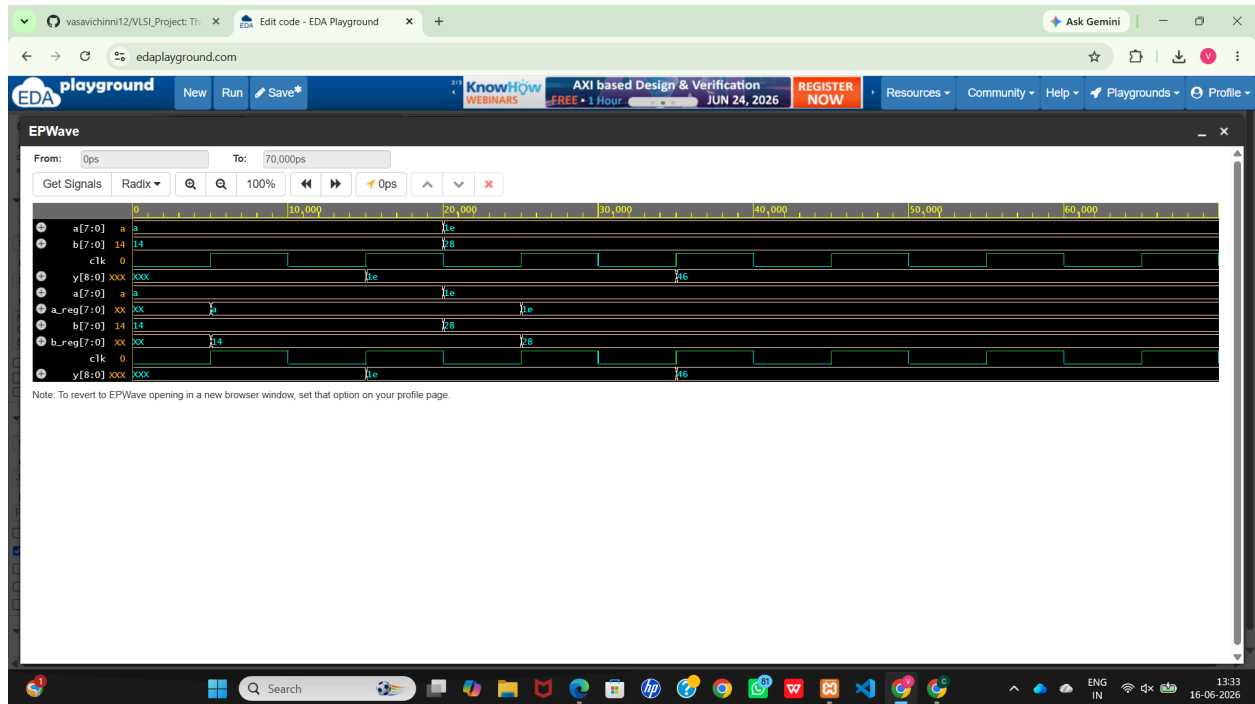
```
    clk = 0;  
    a = 10;  
    b = 20;
```

```
    #20;  
    a = 30;  
    b = 40;
```

```
    #50;  
    $finish;  
end
```

```
endmodule
```

Results:



Conclusion:

The project helped in understanding timing analysis of digital circuits using Verilog in EDA Playground. It demonstrated how timing issues occur and how they can be improved using optimization techniques to achieve correct and efficient design.